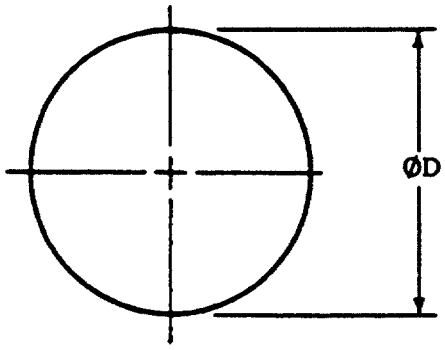
MILITARY SPECIFICATION SHEET
BALLS, BEARING, AIRCRAFT QUALITY STEEL, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bearing described herein shall consist of this specification and the latest issue of DOD-B-82737.



NOTES:

1. All dimensions are in millimeters. Column headings in tolerance tables are in accordance with AFBMA-STD-10.
2. For tolerances by grade see Table II for individual balls and Table III for lots of balls. Tolerances are in micrometers (μm).
3. Grade 10 balls are preferred in most military applications.
4. Referenced government (or nongovernment) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

FIGURE 1. BALLS, BEARING, AIRCRAFT QUALITY STEEL, METRIC.

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TABLE I. Dash numbers and dimensions.

ØD NOM	GRADE 3	GRADE 5	GRADE 10	GRADE 16	GRADE 24	GRADE 48
	DASH NUMBERS					
0.8	308	508	1008	1608	2408	4808
1	310	510	1010	1610	2410	4810
1.2	312	512	1012	1612	2412	4812
1.5	315	515	1015	1615	2415	4815
2	320	520	1020	1620	2420	4820
2.5	325	525	1025	1625	2425	4825
3	330	530	1030	1630	2430	4830
3.5	335	535	1035	1635	2435	4835
4	340	540	1040	1640	2440	4840
4.5	345	545	1045	1645	2445	4845
5	350	550	1050	1650	2450	4850
5.5	355	555	1055	1655	2455	4855
6	360	560	1060	1660	2460	4860
6.5	365	565	1065	1665	2465	4865
7	370	570	1070	1670	2470	4870
7.5	375	575	1075	1675	2475	4875
8	380	580	1080	1680	2480	4880
8.5	385	585	1085	1685	2485	4885
9	390	590	1090	1690	2490	4890
10	3100	5100	10100	16100	24100	48100
11	3110	5110	10110	16110	24110	48110
11.5	3115	5115	10115	16115	24115	48115
12	3120	5120	10120	16120	24120	48120
13	—	—	—	—	24130	48130
14	—	—	—	—	24140	48140
15	—	—	—	—	24150	48150
16	—	—	—	—	24160	48160
17	—	—	—	—	24170	48170
18	—	—	—	—	24180	48180
19	—	—	—	—	24190	48190
20	—	—	—	—	24200	48200
21	—	—	—	—	24210	48210
22	—	—	—	—	24220	48220
23	—	—	—	—	24230	48230
24	—	—	—	—	24240	48240
25	—	—	—	—	24250	48250
26	—	—	—	—	24260	48260
28	—	—	—	—	24280	48280
30	—	—	—	—	24300	48300
32	—	—	—	—	24320	48320
34	—	—	—	—	24340	48340
35	—	—	—	—	24350	48350
36	—	—	—	—	24360	48360
38	—	—	—	—	24380	48380
40	—	—	—	—	24400	48400

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REQUIREMENTS:

1. Material: Steel, aircraft quality conforming to the chemical composition of UNS T11350 in accordance with AMS 6491A and as specified in procurement document.

2. Hardness: Shall be 61-64 HRC or equivalent.

3. Surface Roughness: Tolerance limits as specified in Table II and in accordance with ANSI B46.1.

4. Material Density: Shall be 7.723 grams per cubic centimeter.

5. Part Number: Consists of the letter "D", the basic number of this specification sheet and a dash number taken from Table I.

Example: D82737/1-315

- Dash number
- Specification sheet number

D82737/1-315 Indicates a grade 3 bearing ball of aircraft-quality steel with a basic diameter of 1.5 millimeters.

TABLE II. Tolerances by grade for individual balls.

GRADE	ALLOWABLE BALL DIAMETER VARIATION	ALLOWABLE DEVIATION FROM SPHERICAL FORM	MAXIMUM SURFACE ROUGHNESS ARITHMETICAL AVERAGE
	V_D 1/	W 1/	
3	0.08	0.08	0.012
5	0.13	0.13	0.02
10	0.25	0.25	0.025
16	0.4	0.4	0.025
24	0.6	0.6	0.05
48	1.2	1.2	0.08

1/ Tolerances in micrometers (μm)

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TABLE III. Tolerances by grade for lots of balls.

GRADE	ALLOWABLE LOT DIAMETER VARIATION [1/]	BASIC DIAMETER TOLERANCE [1/]	ALLOWABLE BALL GAGE DEVIATION [1/]		CONTAINER MARKING INCREMENT [1/]
			HIGH	LOW	
3	0.13	±0.75	±0.75	−0.75	0.25
5	0.25	+1.25	+1.25	−1	0.25
10	0.5	±2.5	+1.25	−1	0.25
16	0.8	±2.5	+1.25	−1	0.25
24	1.2	±2.5	+2.5	−2.5	0.25
48	2.4	±5	—[2/]	—[2/]	1.25

[1/] Tolerances in micrometers (μm)
[2/] Not applicable

Custodians:
Army - AT
Air Force - 99

Review activities:
Navy - SH
Air Force - 84
DLA - IS

Preparing activity:
Navy - OS

Agent:
DLA - IS

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